

Introduction to LangSmith

Instructor

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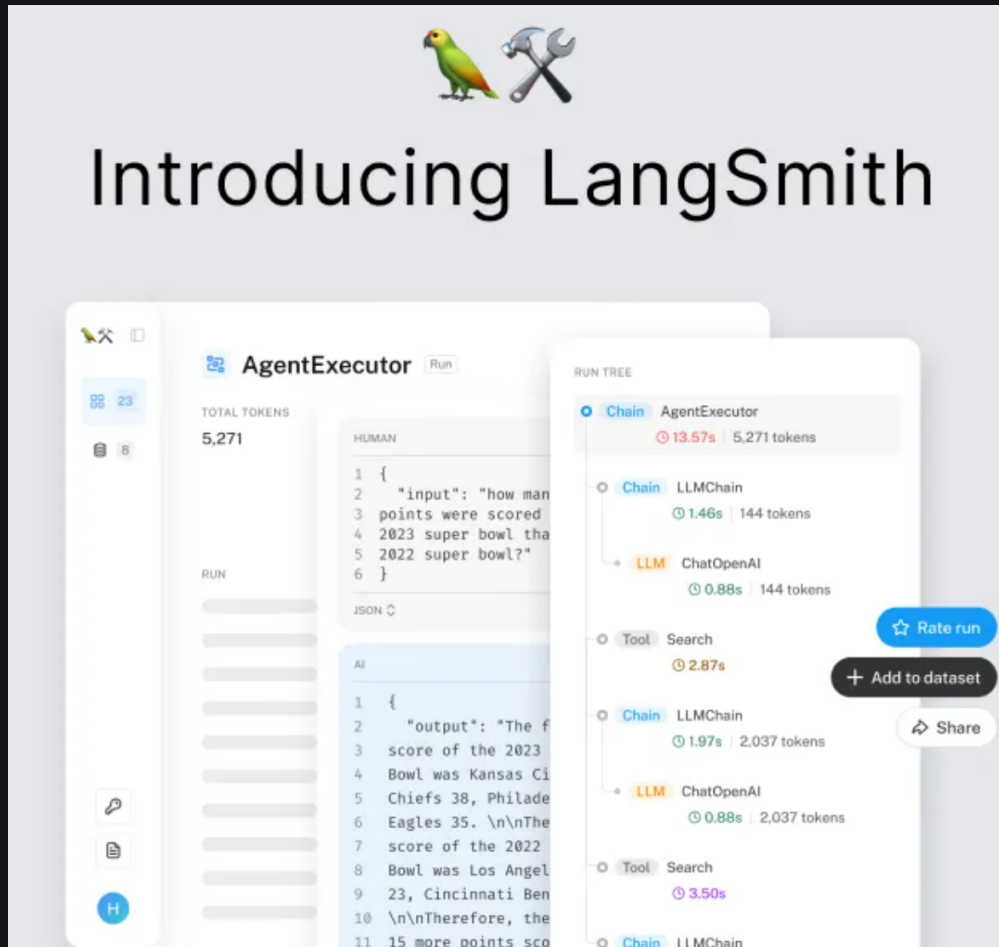
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Understanding LangSmith



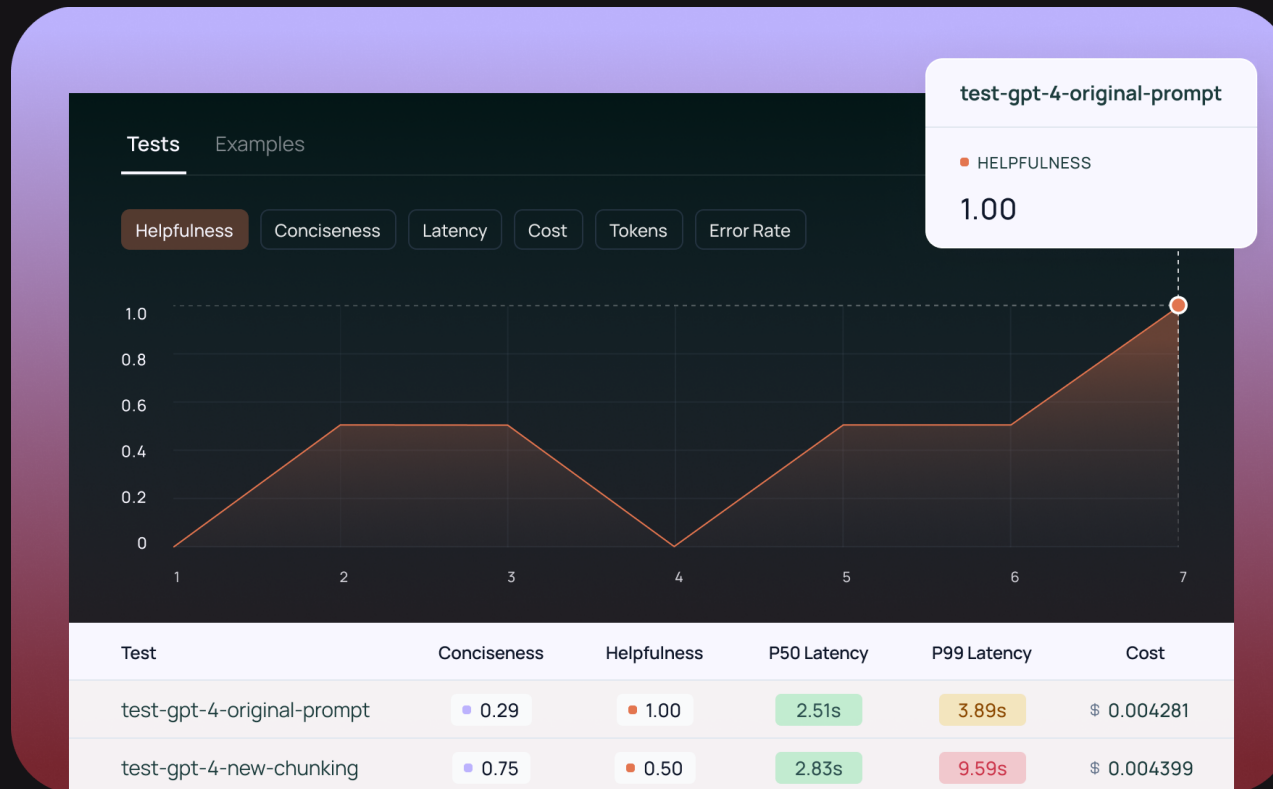
- A unified platform from LangChain, for debugging, testing, evaluating, and monitoring LLM applications
- Helps to easily track changes, debug, run and evaluate any LLM application and agents

Developed with more visibility

The screenshot displays the LangSmith interface for a chat application. On the left, a 'TRACE' sidebar shows a sequence of steps: 'chat-langchain' (5.13s, 5,846 tokens), 'FindDocs' (0.70s), 'RetrievalChainWith...' (0.68s), 'Retriever' (0.68s, highlighted), 'GenerateResponse' (5.07s), and 'ChatOpenAI' (4.33s). The main panel shows the details for the 'Retriever' step. The input is 'How do I use a RecursiveUrlLoader to load content from a page?'. The output is a list of documents, including 'Recursive URL | Langchain [Skip to main content](#_docu...', 'langchain_community.document_loader.RecursiveUrlLoader - La...', 'lazy_load() Lazy load web pages. load() Load web pages. load_An...', 'python os.environment["LANGCHAIN_TRACING_V2" = "true" os....', and 'for result in results: print("Content:", result[0].page_content, "--...'. At the bottom, there are buttons for 'Add to Dataset', 'Annotate', and 'Playground'.

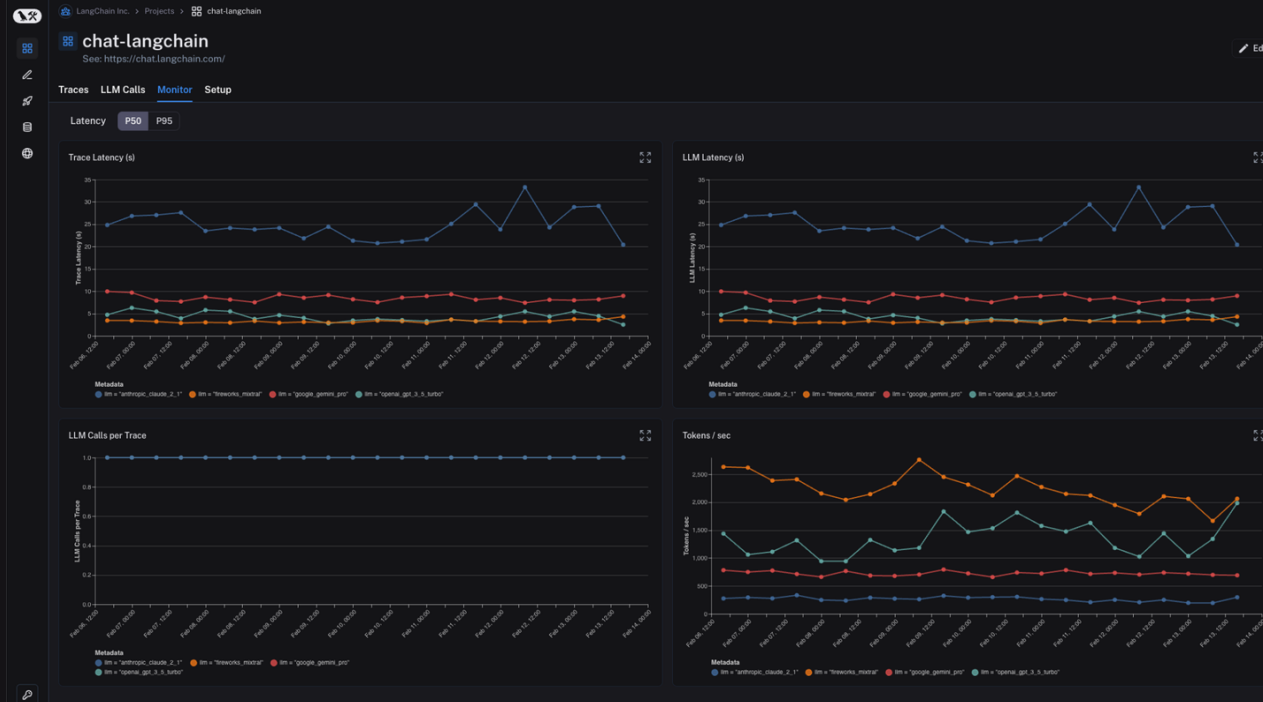
- LangSmith allows to have complete visibility into the entire sequence of calls.
- Help in finding out errors and performance bottlenecks in LLM applications
- Makes it easy to debug and experiment with LLM apps.

Test & Evaluate LLM Apps



- LangSmith allows us to receive human feedback during LLM app runs or utilize AI-assisted evaluation methods.
- The reference LangSmith datasets can be used to test the LLM app.
- Supports both offline and online evaluation to test for problems and drift
- Has many pre-built and custom evaluators that can check for relevance, correctness, harmfulness, and more

Monitoring in Production



- LangSmith provides full visibility into the operations of a production application
- LangSmith offers monitoring charts that enable tracking of key metrics over time
- Charts such as Trace Latency, Tokens/Second, Cost, and Feedback are available for viewing

Thank You
